

Set 7: Customer Demographics Analysis

7A. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

```
library(ggplot2)
```

```
customer <- data.frame(
```

```
  CustomerID = c(1,2,3),
```

```
  Age = c(28,35,42),
```

```
  Gender = c("Female","Male","Female"),
```

```
  Income = c(50000,60000,75000)
```

```
)
```

```
ggplot(customer, aes(x = factor(CustomerID), y = Age, fill = factor(CustomerID))) +
```

```
  geom_col() +
```

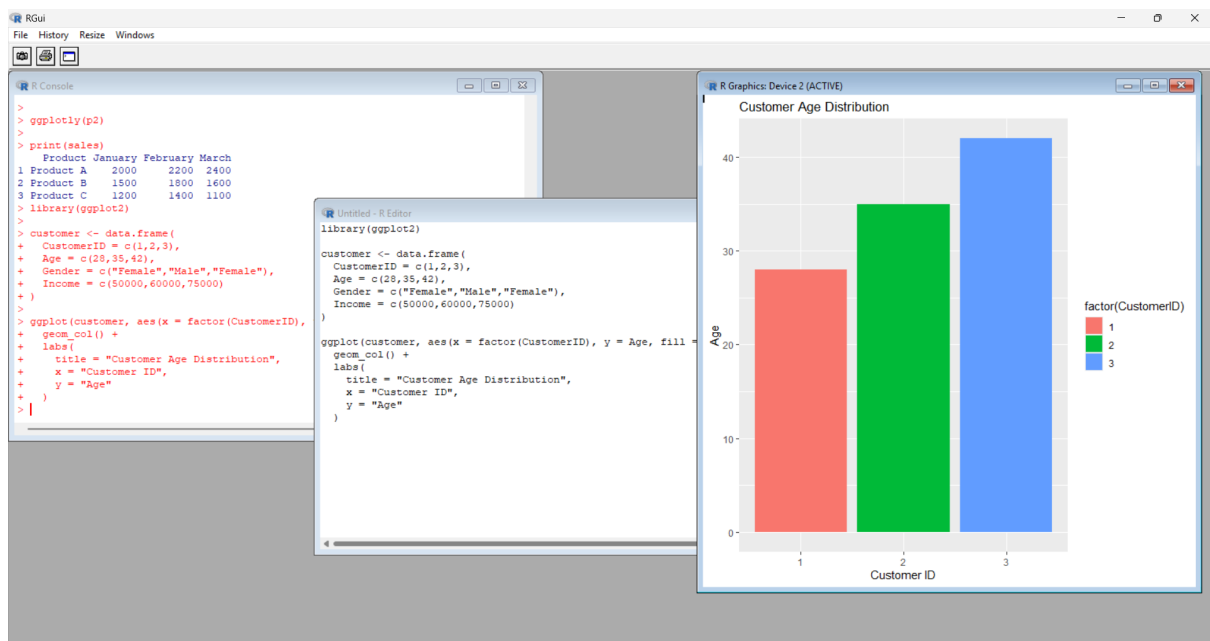
```
  labs(
```

```
    title = "Customer Age Distribution",
```

```
    x = "Customer ID",
```

```
    y = "Age"
```

```
)
```

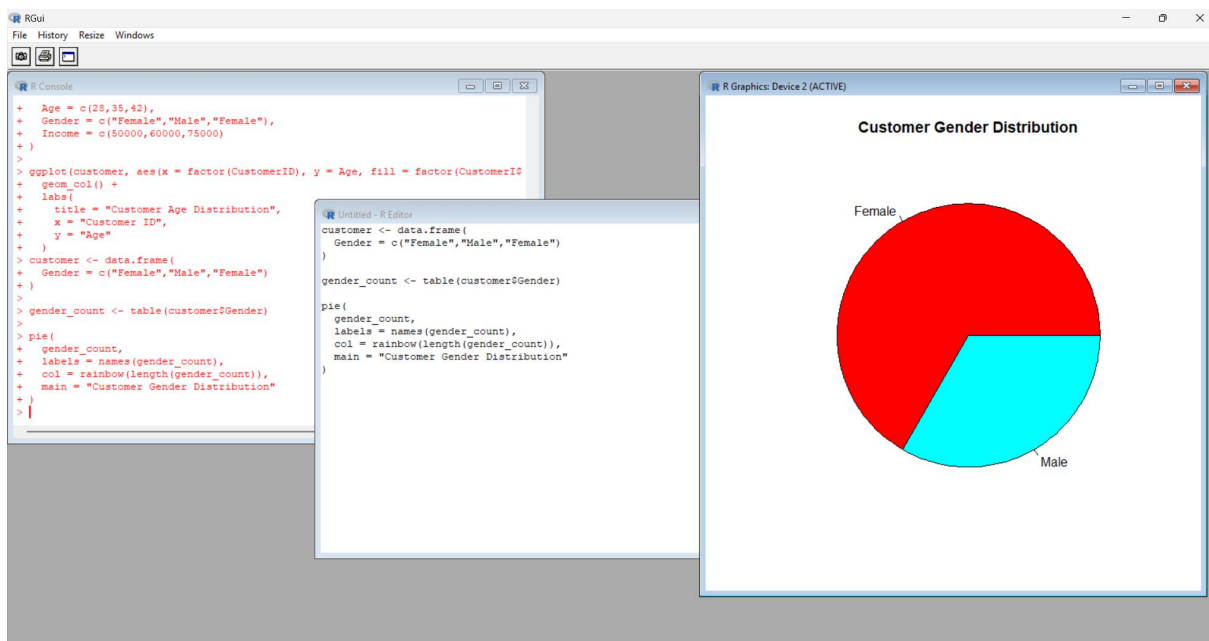


7B. Generate a pie chart to represent the distribution of customers by gender.

```
customer <- data.frame(  
  Gender = c("Female","Male","Female")  
)
```

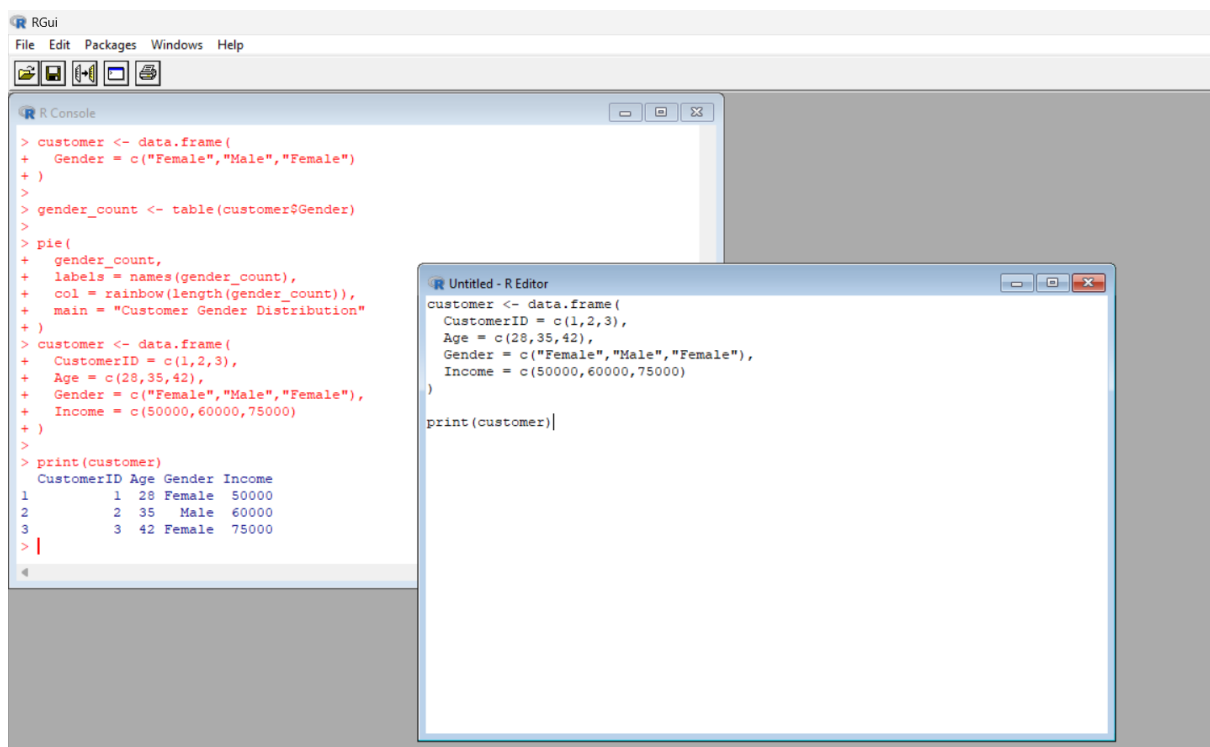
```
gender_count <- table(customer$Gender)
```

```
pie(  
  gender_count,  
  labels = names(gender_count),  
  col = rainbow(length(gender_count)),  
  main = "Customer Gender Distribution"  
)
```



7C. Build a table to show the demographic information for each customer. Label the table elements.

```
customer <- data.frame(  
  CustomerID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)  
  
print(customer)
```



7D. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

```
library(ggplot2)
```

```
library(plotly)
```

```
customer <- data.frame(  
  CustomerID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)
```

```
p1 <- ggplot(customer, aes(x = factor(CustomerID), y = Age, fill = factor(CustomerID))) +  
  geom_col() +  
  labs(  
    title = "Customer Age Distribution",  
    x = "Customer ID",  
    y = "Age"  
  )
```

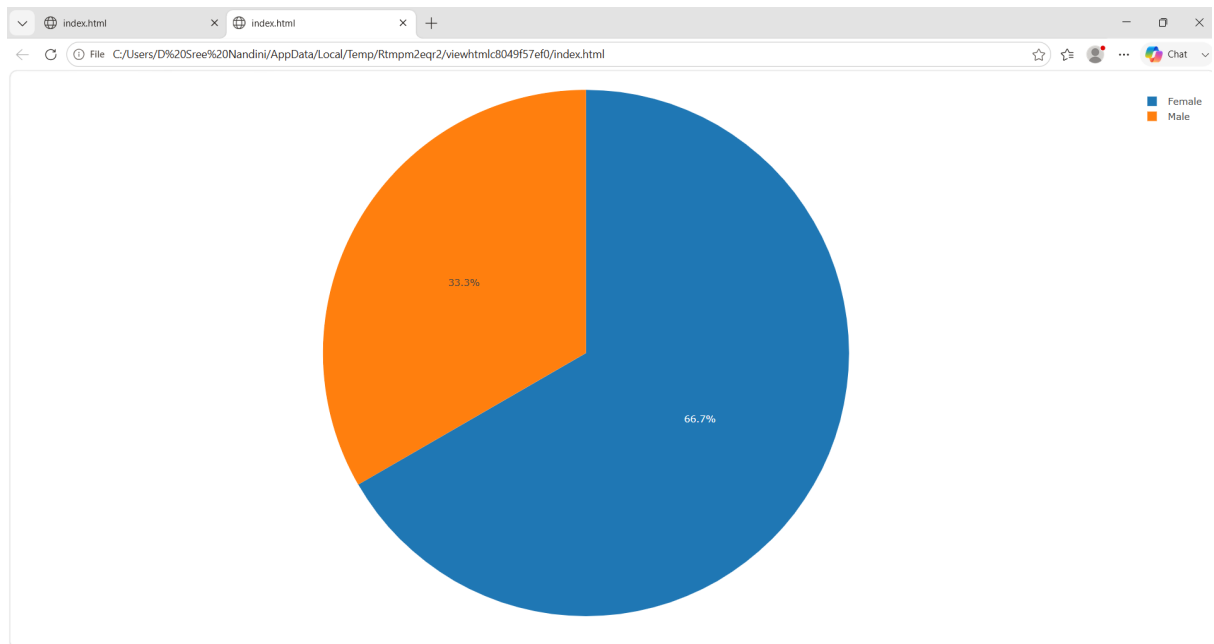
```
ggplotly(p1)
```

```
gender_count <- table(customer$Gender)
```

```
p2 <- plot_ly(  
  labels = names(gender_count),  
  values = as.numeric(gender_count),  
  type = "pie"  
)
```

```
p2
```

```
print(customer)
```



Set 8: Employee Performance Analysis

8A. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```
library(ggplot2)
```

```
employee <- data.frame(
```

```
  EmployeeID = c(1,2,3),
```

```
  Department = c("Sales","HR","Marketing"),
```

```
  YearsService = c(5,3,7),
```

```
  PerformanceScore = c(85,92,78)
```

```
)
```

```
ggplot(employee, aes(x = YearsService, y = PerformanceScore, group = 1)) +
```

```
  geom_line(color = "blue", linewidth = 1) +
```

```
  geom_point(color = "red", size = 3) +
```

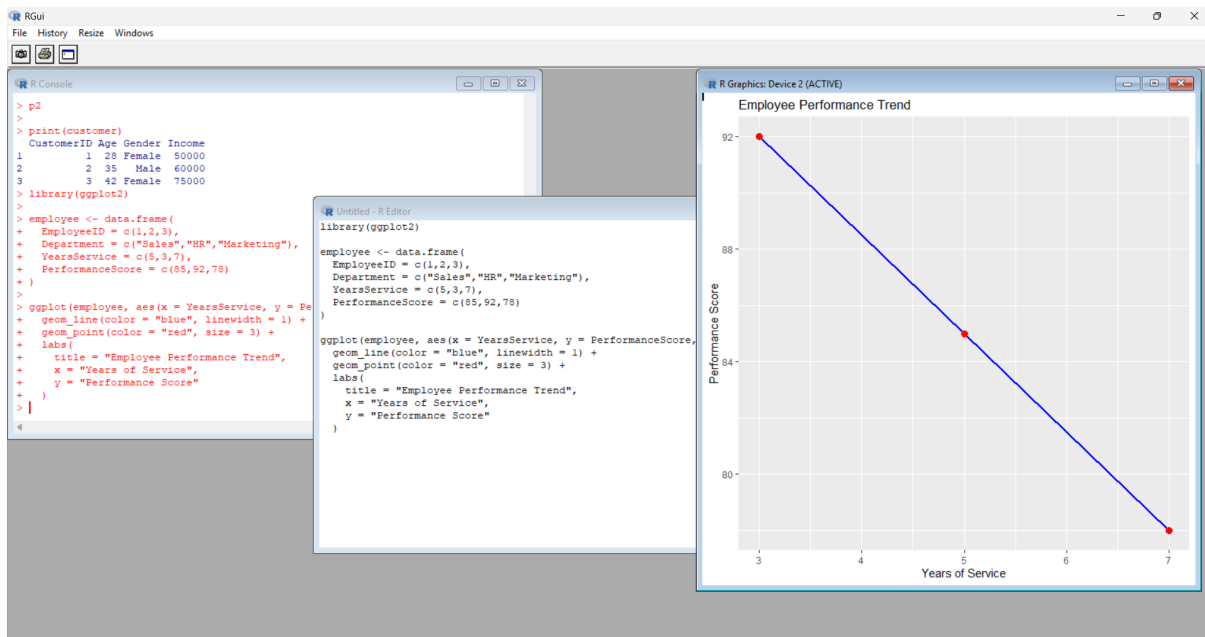
```
  labs(
```

```
    title = "Employee Performance Trend",
```

```
    x = "Years of Service",
```

```
    y = "Performance Score"
```

```
)
```



8B. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
library(ggplot2)
```

```
employee <- data.frame(
```

```
  Department = c("Sales","HR","Marketing")
```

```
)
```

```
ggplot(employee, aes(x = Department, fill = Department)) +
```

```
  geom_bar() +
```

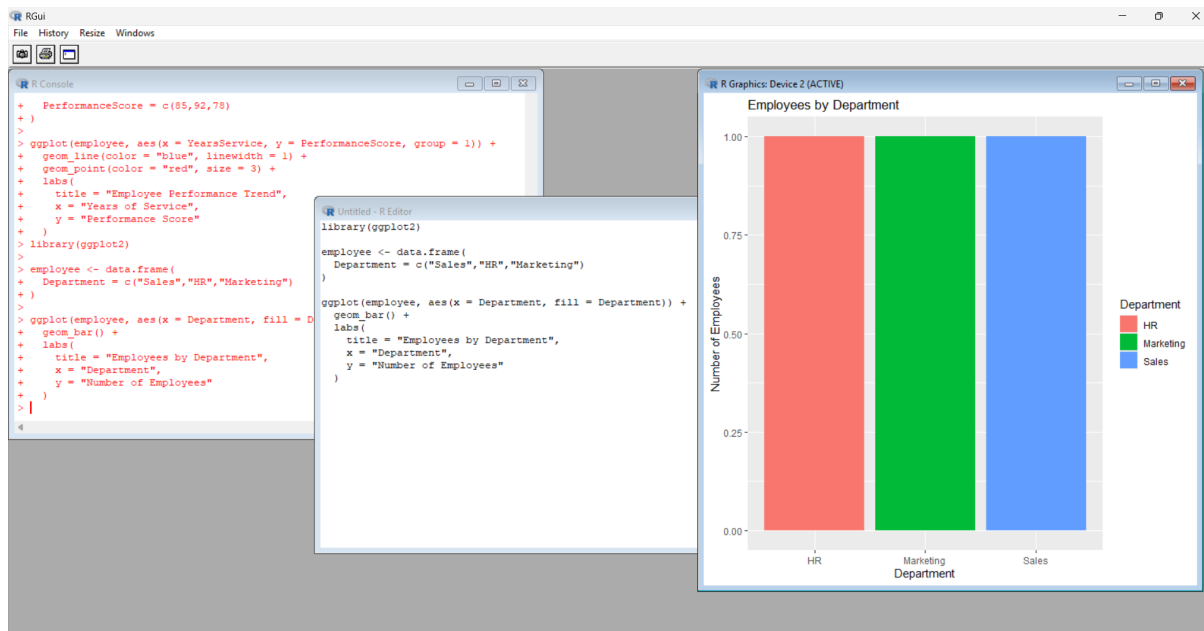
```
  labs(
```

```
    title = "Employees by Department",
```

```
    x = "Department",
```

```
    y = "Number of Employees"
```

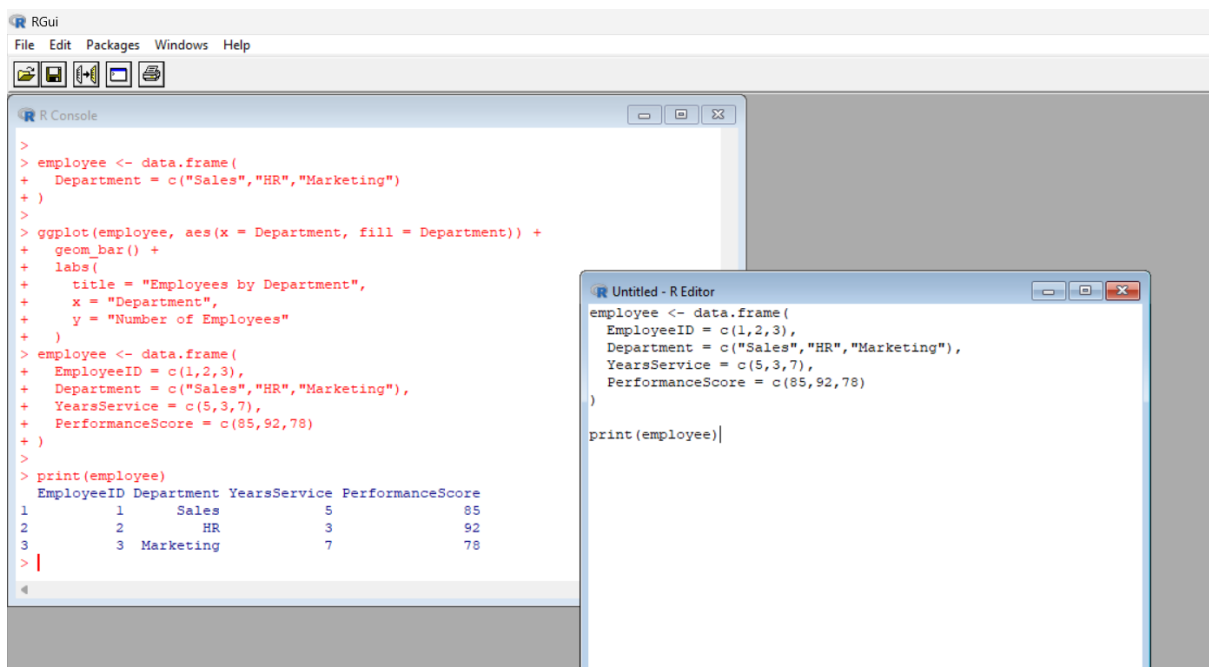
```
)
```



8C. Build a table to display the performance data for each employee. Label the table elements.

```
employee <- data.frame(  
  EmployeeID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  YearsService = c(5,3,7),  
  PerformanceScore = c(85,92,78)  
)
```

```
print(employee)
```



8D. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

```
library(ggplot2)
```

```
library(plotly)
```

```
employee <- data.frame(  
  EmployeeID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  YearsService = c(5,3,7),  
  PerformanceScore = c(85,92,78)  
)
```

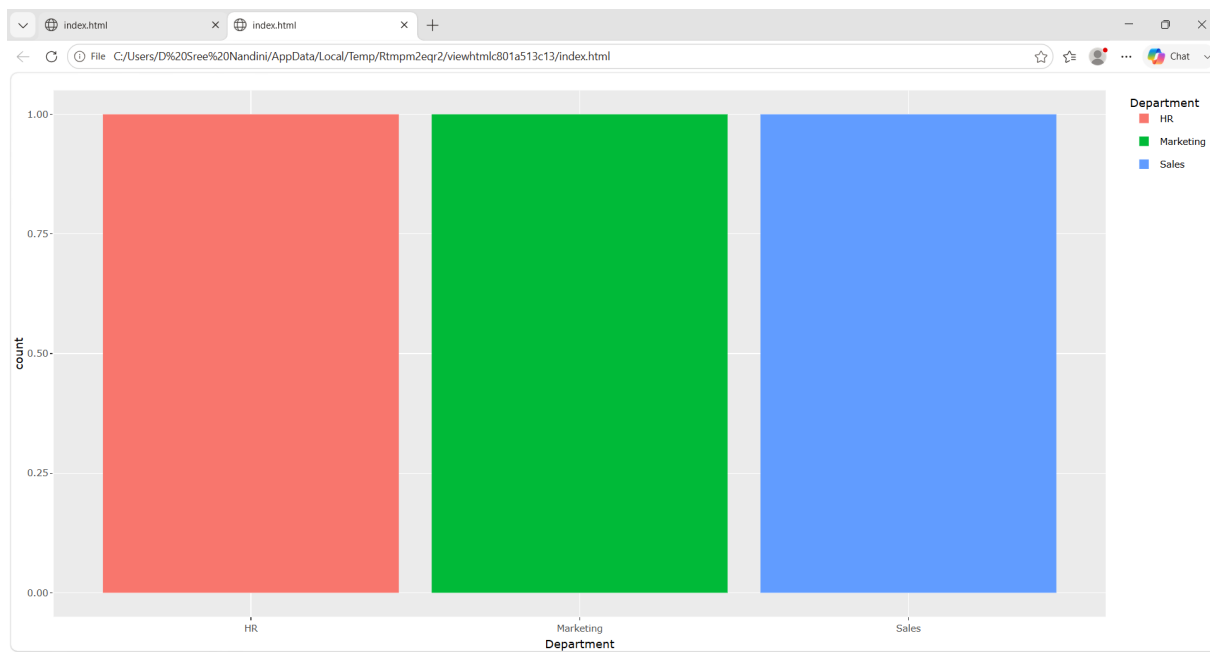
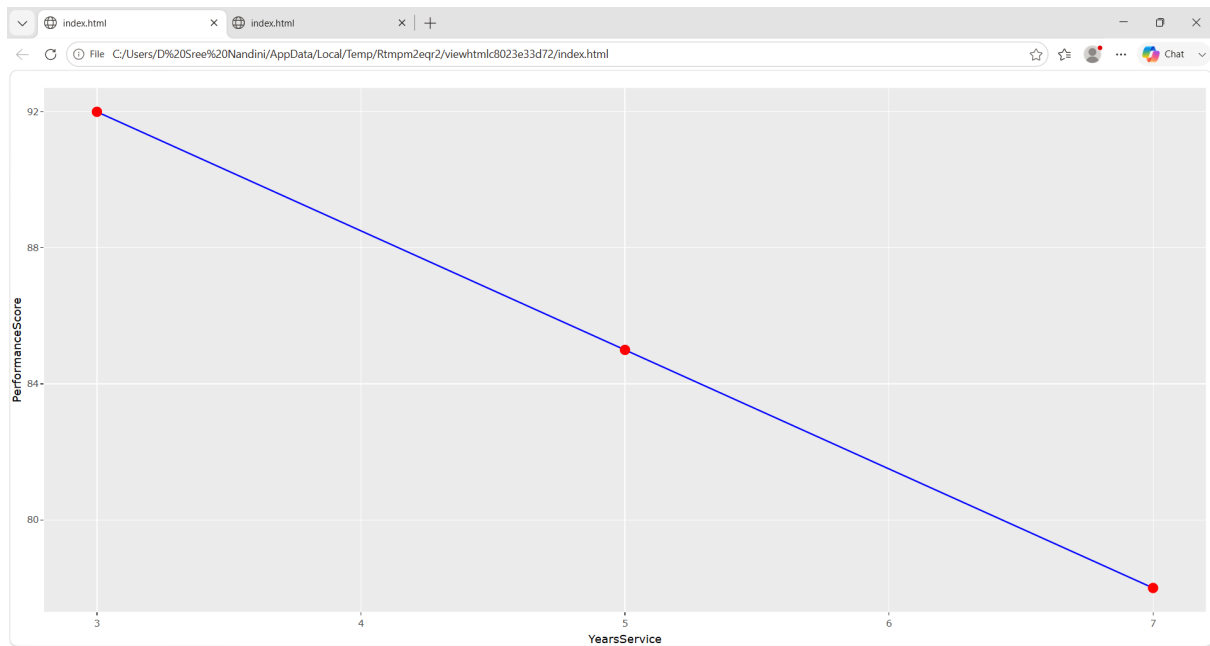
```
p1 <- ggplot(employee, aes(x = YearsService, y = PerformanceScore, group = 1)) +  
  geom_line(color = "blue") +  
  geom_point(color = "red", size = 3)
```

```
ggplotly(p1)
```

```
p2 <- ggplot(employee, aes(x = Department, fill = Department)) +  
  geom_bar()
```

```
ggplotly(p2)
```

```
print(employee)
```



Set 9:

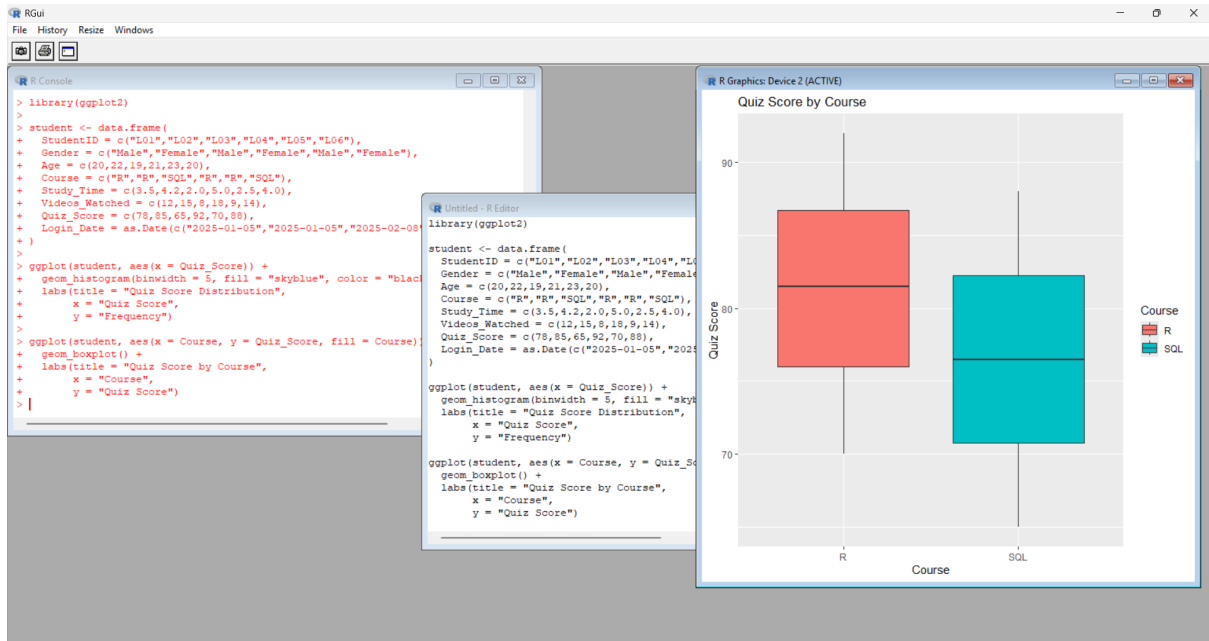
9A. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

```
library(ggplot2)
```

```
student <- data.frame(  
  StudentID = c("L01","L02","L03","L04","L05","L06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Age = c(20,22,19,21,23,20),  
  Course = c("R","R","SQL","R","R","SQL"),  
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),  
  Videos_Watched = c(12,15,8,18,9,14),  
  Quiz_Score = c(78,85,65,92,70,88),  
  Login_Date = as.Date(c("2025-01-05","2025-01-05","2025-02-08","2025-02-08","2025-03-12","2025-03-12"))  
)
```

```
ggplot(student, aes(x = Quiz_Score)) +  
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +  
  labs(title = "Quiz Score Distribution",  
       x = "Quiz Score",  
       y = "Frequency")
```

```
ggplot(student, aes(x = Course, y = Quiz_Score, fill = Course)) +  
  geom_boxplot() +  
  labs(title = "Quiz Score by Course",  
       x = "Course",  
       y = "Quiz Score")
```

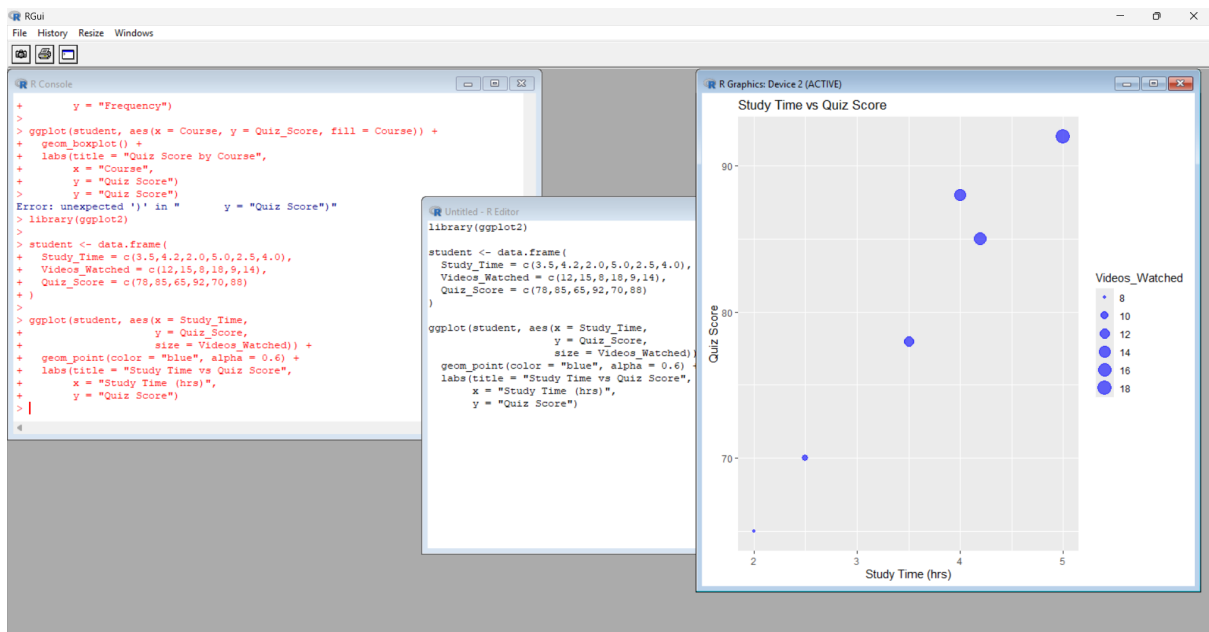


9B. Create a scatter plot using R programming between Study Time and Quiz Score. Represent Videos Watched using bubble size. Apply transparency. Explain the learning pattern.

```
library(ggplot2)
```

```
student <- data.frame(  
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),  
  Videos_Watched = c(12,15,8,18,9,14),  
  Quiz_Score = c(78,85,65,92,70,88)  
)
```

```
ggplot(student, aes(x = Study_Time,  
  y = Quiz_Score,  
  size = Videos_Watched)) +  
  geom_point(color = "blue", alpha = 0.6) +  
  labs(title = "Study Time vs Quiz Score",  
  x = "Study Time (hrs)",  
  y = "Quiz Score")
```



9C. Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

```
library(ggplot2)
```

```
student <- data.frame(
```

```
  Login_Date = as.Date(c("2025-01-05","2025-01-05","2025-02-08","2025-02-08","2025-03-12","2025-03-12")),
```

```
  Quiz_Score = c(78,85,65,92,70,88)
```

```
)
```

```
student$Month <- format(student$Login_Date,"%Y-%m")
```

```
avg <- aggregate(Quiz_Score ~ Month, data = student, mean)
```

```
ggplot(avg, aes(x = Month, y = Quiz_Score, group = 1)) +
```

```
  geom_line(color = "blue") +
```

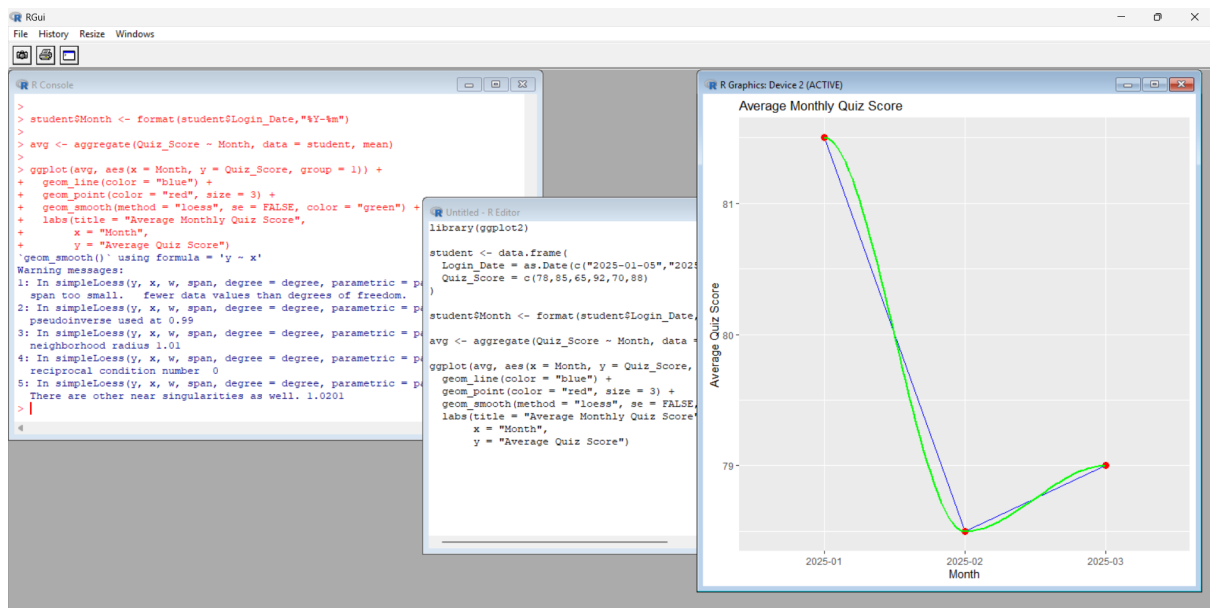
```
  geom_point(color = "red", size = 3) +
```

```
  geom_smooth(method = "loess", se = FALSE, color = "green") +
```

```
  labs(title = "Average Monthly Quiz Score",
```

```
    x = "Month",
```

```
    y = "Average Quiz Score")
```



9D. Import Online_Learning_Activity.csv into Tableau. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini-interactive dashboard.

```
library(ggplot2)
```

```
library(plotly)
```

```
student <- data.frame(  
  Course = c("R","R","SQL","R","R","SQL"),  
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),  
  Quiz_Score = c(78,85,65,92,70,88),  
  Gender = c("Male","Female","Male","Female","Male","Female")  
)
```

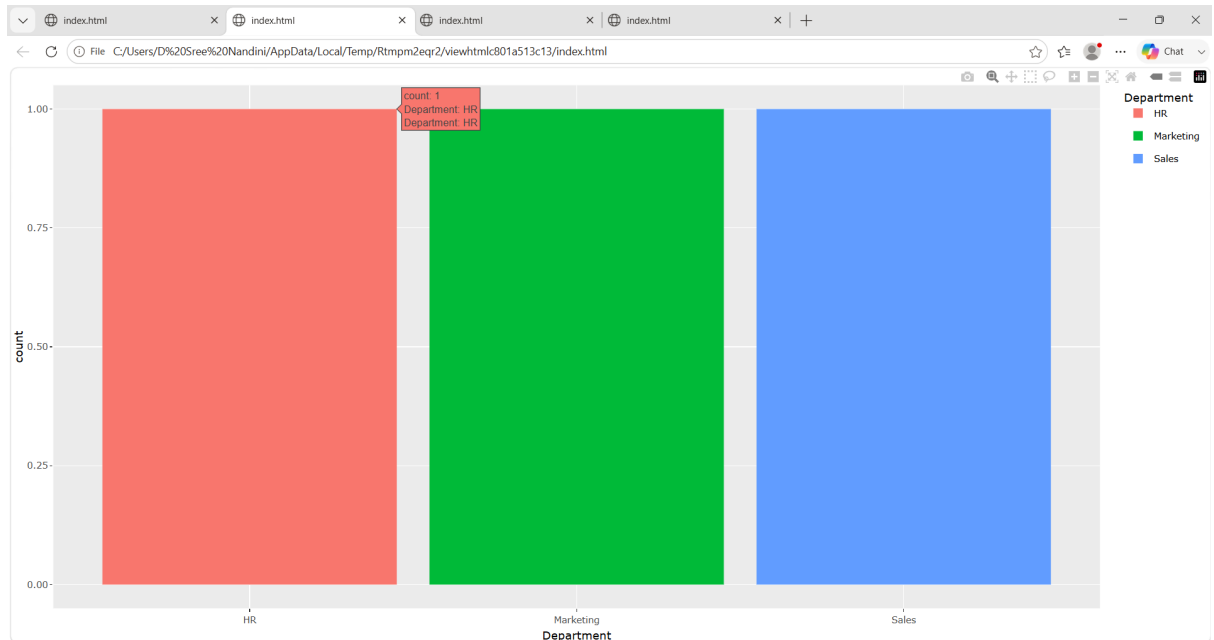
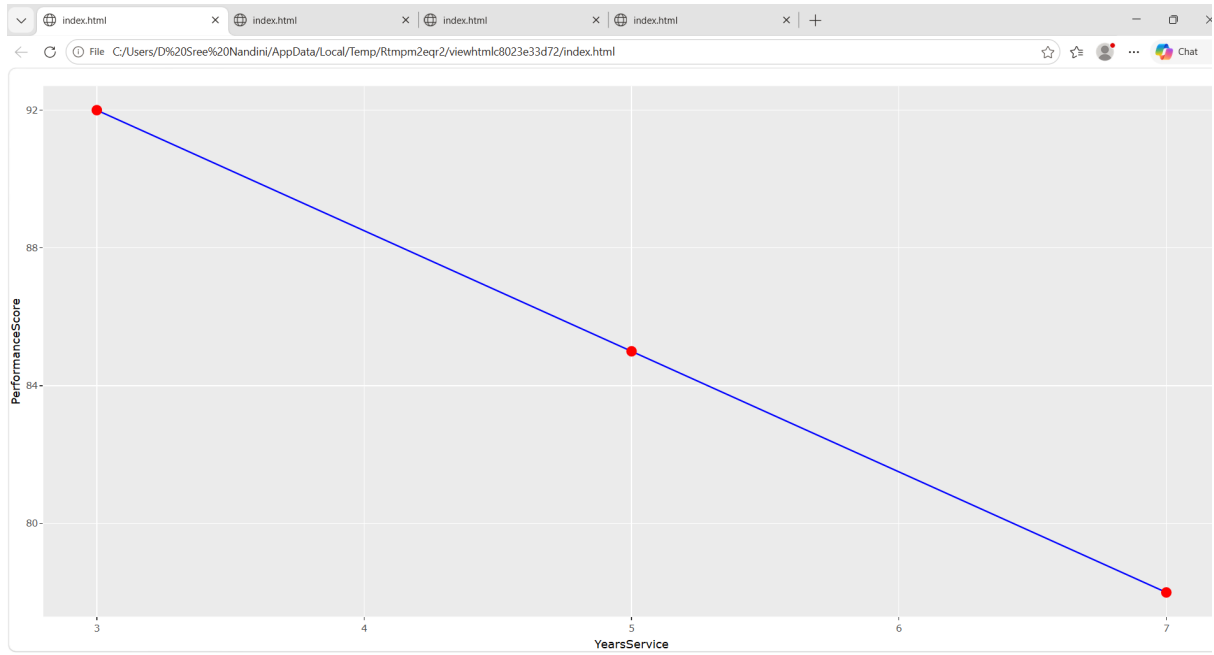
```
avg <- aggregate(Quiz_Score ~ Course, data = student, mean)
```

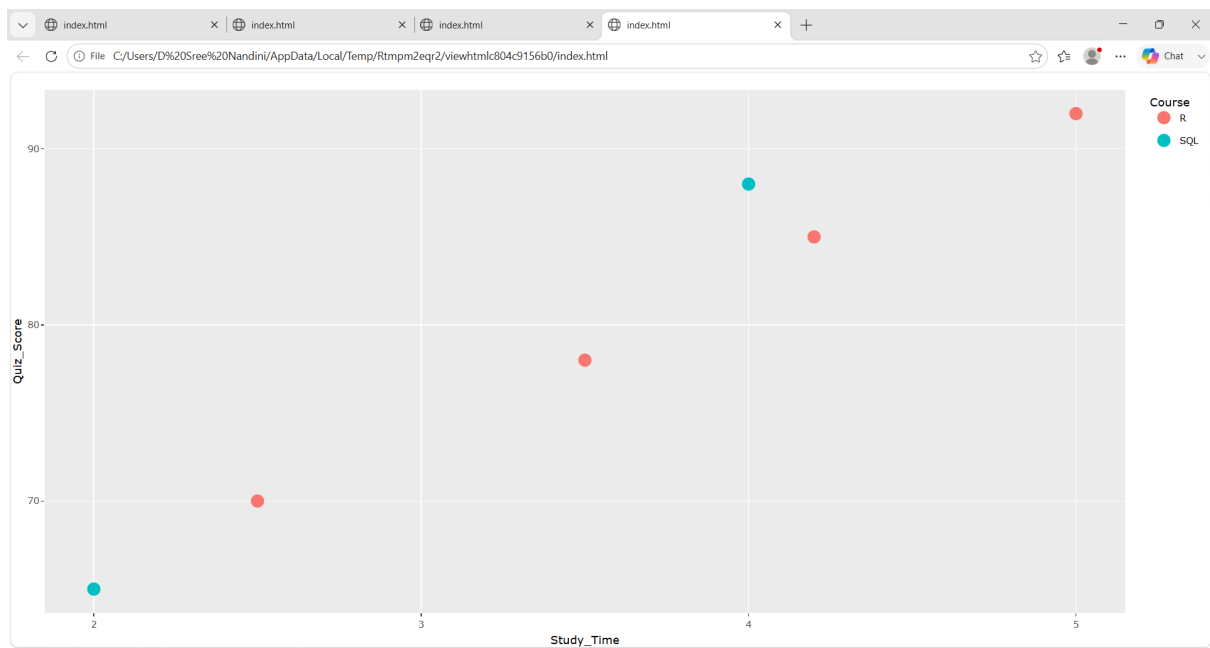
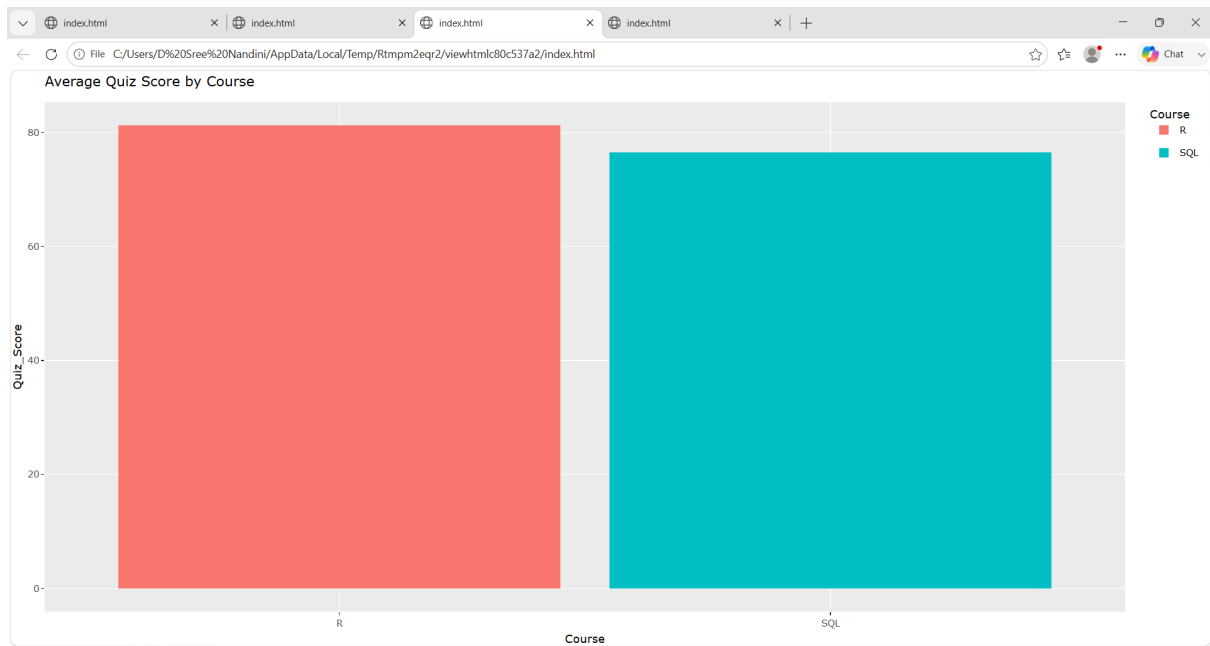
```
p1 <- ggplot(avg, aes(x = Course, y = Quiz_Score, fill = Course)) +  
  geom_col() +  
  labs(title = "Average Quiz Score by Course")
```

```
ggplotly(p1)
```

```
p2 <- ggplot(student, aes(x = Study_Time,  
  y = Quiz_Score,  
  color = Course)) +  
  geom_point(size = 4)
```

```
ggplotly(p2)
```





Set 10: Survey Responses Analysis

10A. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

```
library(ggplot2)

survey <- data.frame(

  SurveyID = c(1,2,3),

  Question1 = c("A","B","C"),

  Question2 = c("B","A","A"),

  Question3 = c("C","D","B")

)

q1 <- as.data.frame(table(survey$Question1))

colnames(q1) <- c("Response","Count")

ggplot(q1, aes(x = Response, y = Count, fill = Response)) +

  geom_col() +

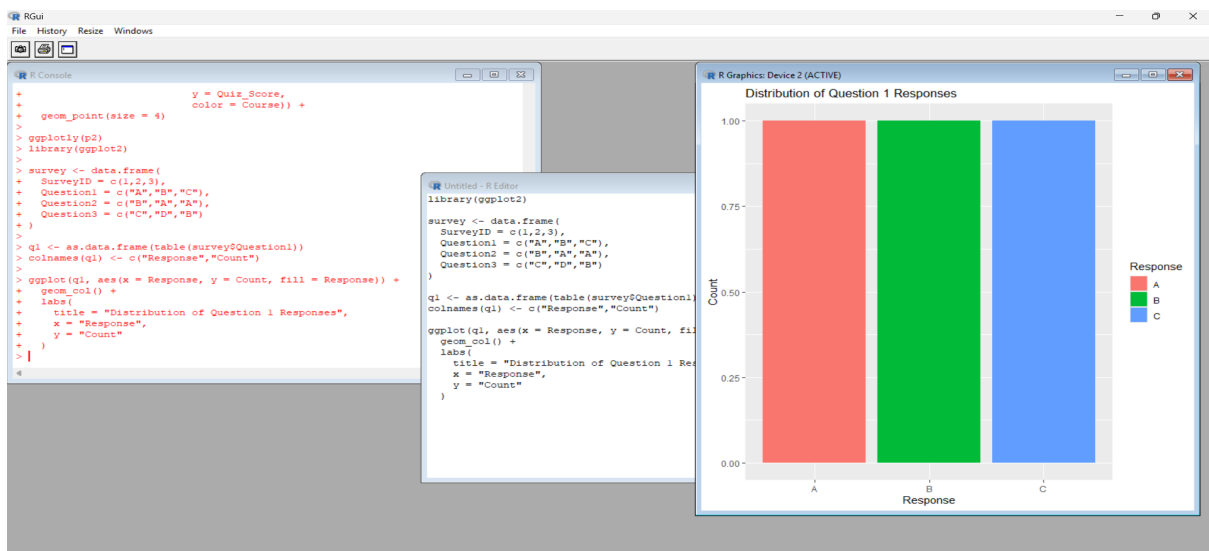
  labs(

    title = "Distribution of Question 1 Responses",

    x = "Response",

    y = "Count"

  )
```



10B. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
library(ggplot2)
```

```
survey <- data.frame(
```

```
  SurveyID = c(1,2,3),
```

```
  Question1 = c("A","B","C"),
```

```
  Question2 = c("B","A","A"),
```

```
  Question3 = c("C","D","B")
```

```
)
```

```
response <- data.frame(
```

```
  Question = rep(c("Question1","Question2","Question3"), each = 3),
```

```
  Answer = c(survey$Question1, survey$Question2, survey$Question3)
```

```
)
```

```
ggplot(response, aes(x = Question, fill = Answer)) +
```

```
  geom_bar() +
```

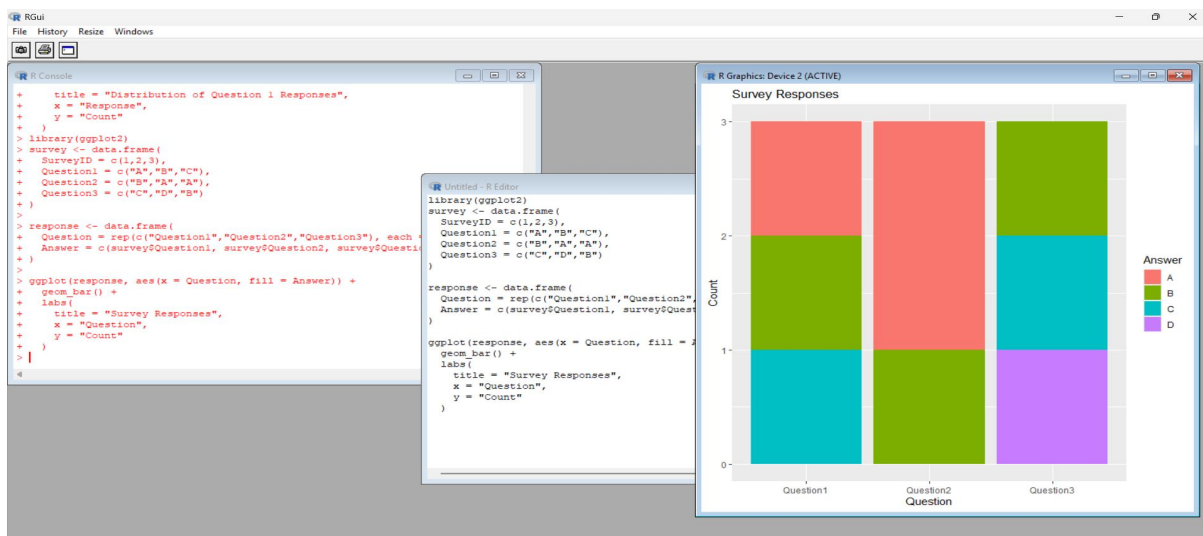
```
  labs(
```

```
    title = "Survey Responses",
```

```
    x = "Question",
```

```
    y = "Count"
```

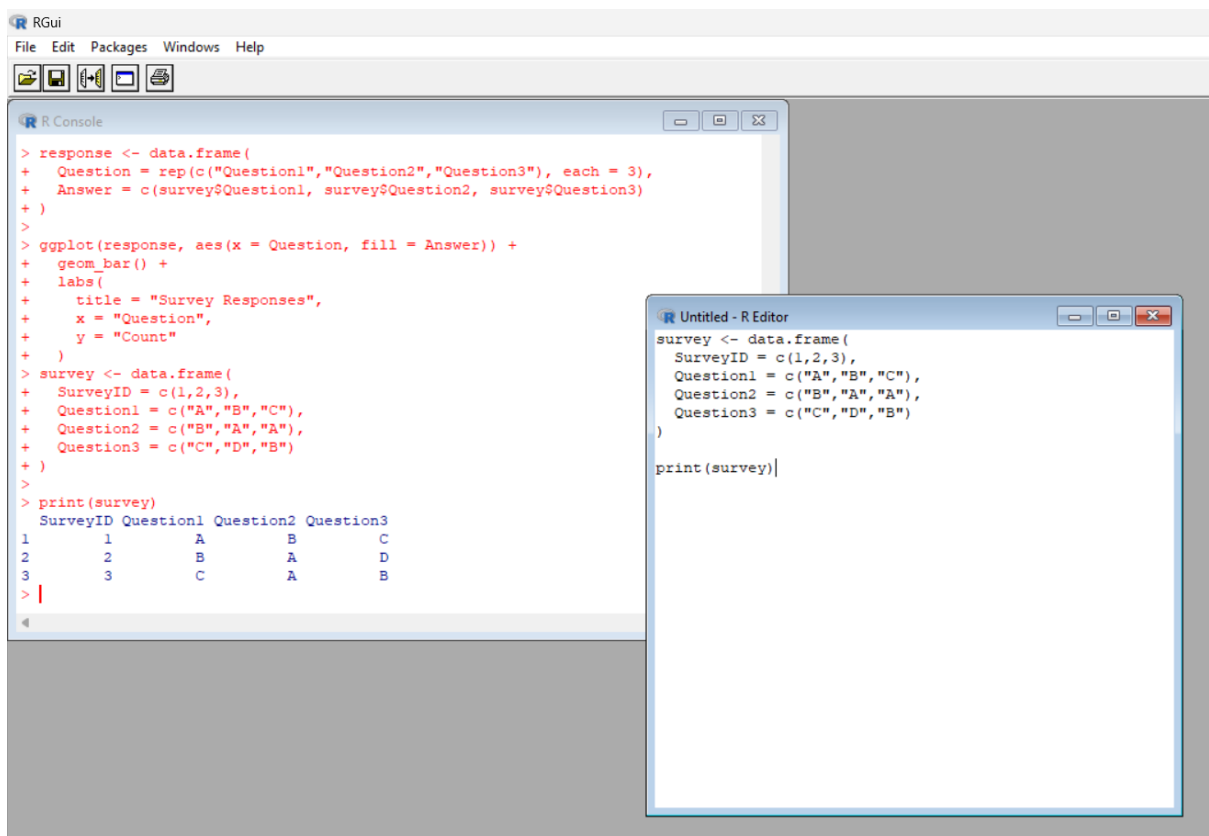
```
)
```



10C. Build a table to show the survey response data for each survey. Label the table elements.

```
survey <- data.frame(  
  SurveyID = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)
```

```
print(survey)
```



10D. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

```
library(ggplot2)
```

```
library(plotly)
```

```
survey <- data.frame(  
  SurveyID = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)
```

```
q1 <- as.data.frame(table(survey$Question1))  
colnames(q1) <- c("Response","Count")
```

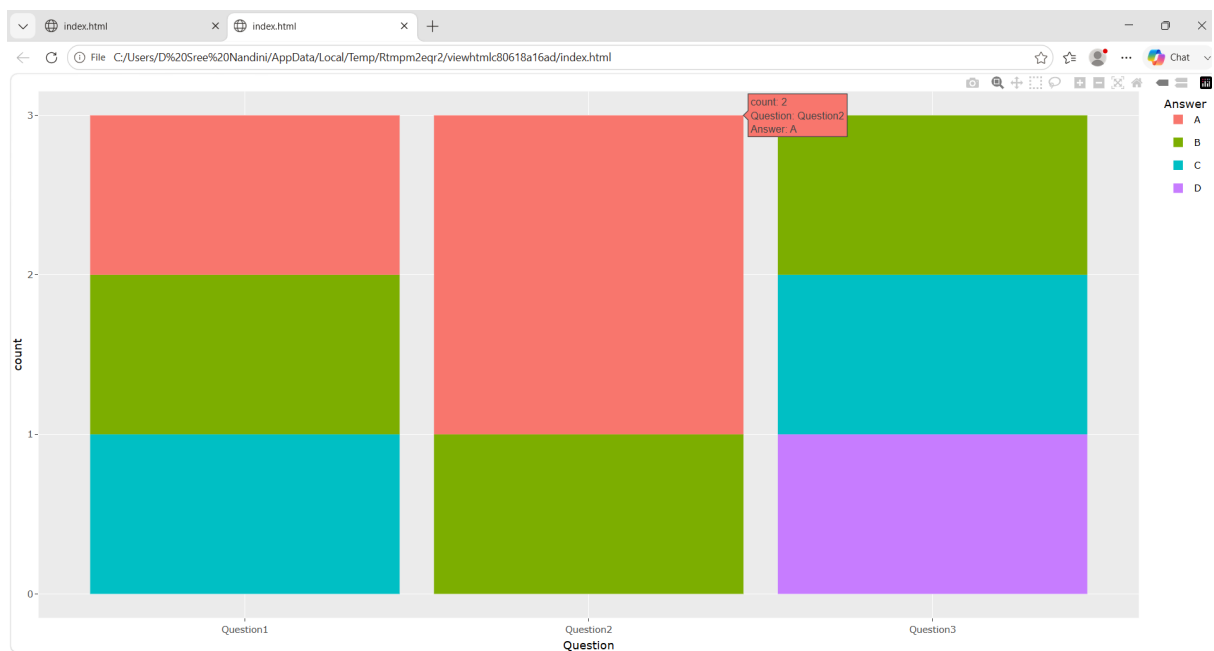
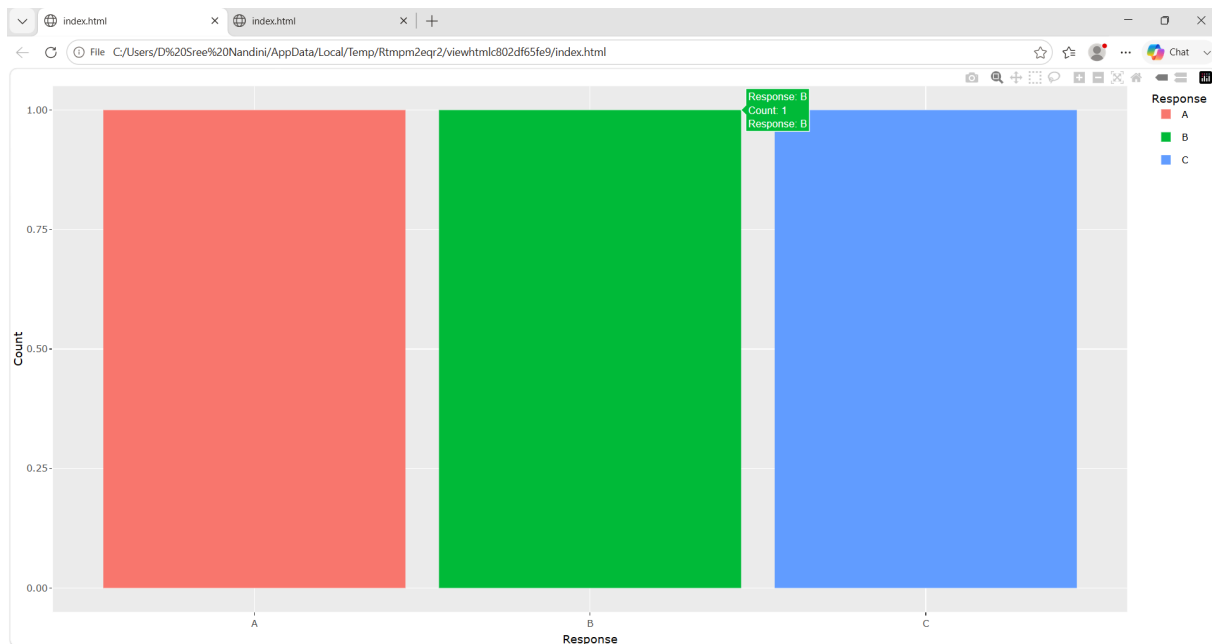
```
p1 <- ggplot(q1, aes(x = Response, y = Count, fill = Response)) +  
  geom_col()
```

```
ggplotly(p1)
```

```
response <- data.frame(  
  Question = rep(c("Question1","Question2","Question3"), each = 3),  
  Answer = c(survey$Question1, survey$Question2, survey$Question3)  
)
```

```
p2 <- ggplot(response, aes(x = Question, fill = Answer)) +  
  geom_bar()  
ggplotly(p2)
```

```
print(survey)
```



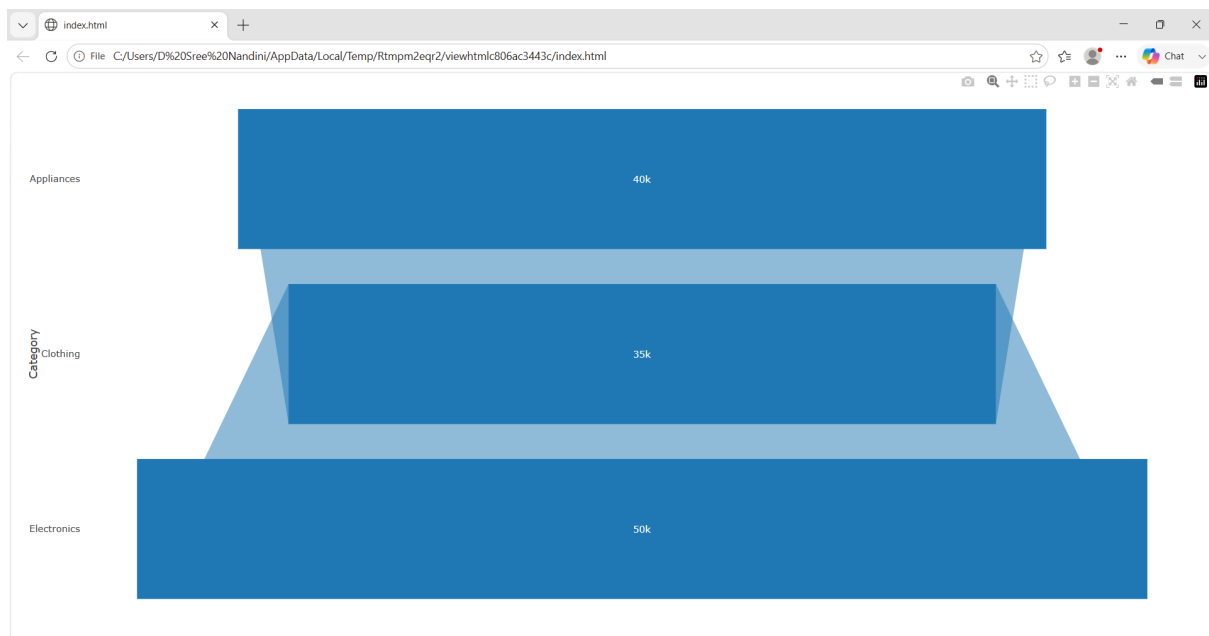
Set 11:Product Category Analysis

11A.Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

```
library(plotly)
```

```
product <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)
```

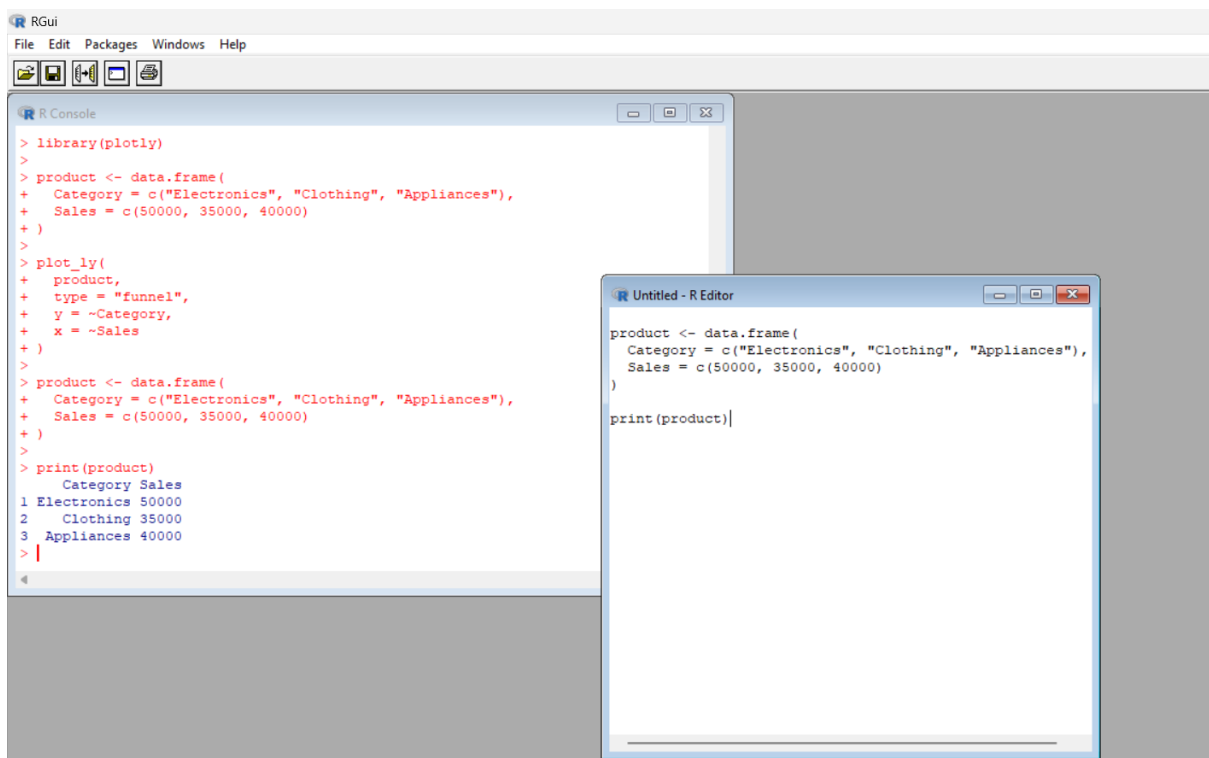
```
plot_ly(  
  product,  
  type = "funnel",  
  y = ~Category,  
  x = ~Sales  
)
```



11B. Build a table to display the sales data for each product category. Label the table elements.

```
product <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)
```

```
print(product)
```



11C. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

```
library(plotly)

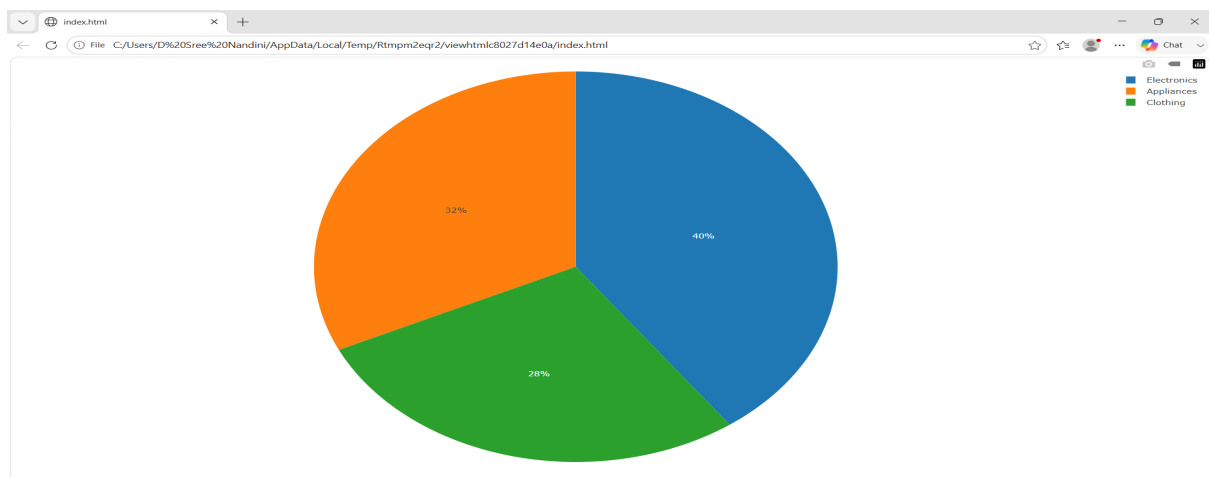
product <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)

p1 <- plot_ly(
  product,
  type = "funnel",
  y = ~Category,
  x = ~Sales
)

p1

p2 <- plot_ly(
  product,
  labels = ~Category,
  values = ~Sales,
  type = "pie"
)

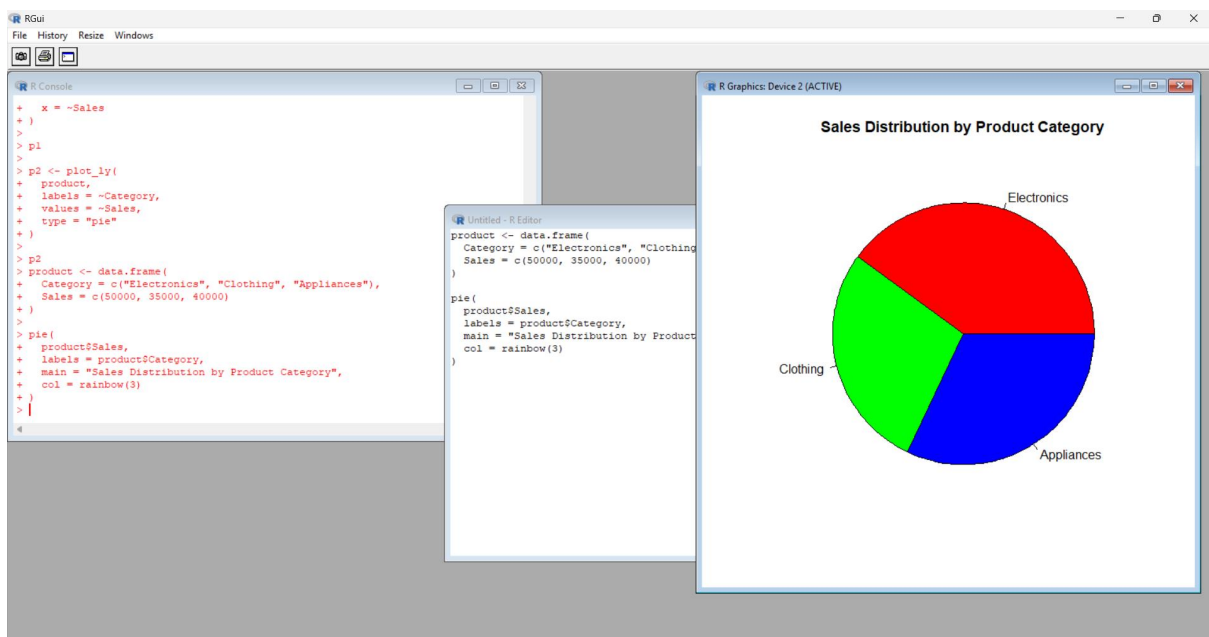
p2
```



11D. Create a pie chart to represent the distribution of sales across product categories. Include labels.

```
product <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)
```

```
pie(  
  product$Sales,  
  labels = product$Category,  
  main = "Sales Distribution by Product Category",  
  col = rainbow(3)  
)
```



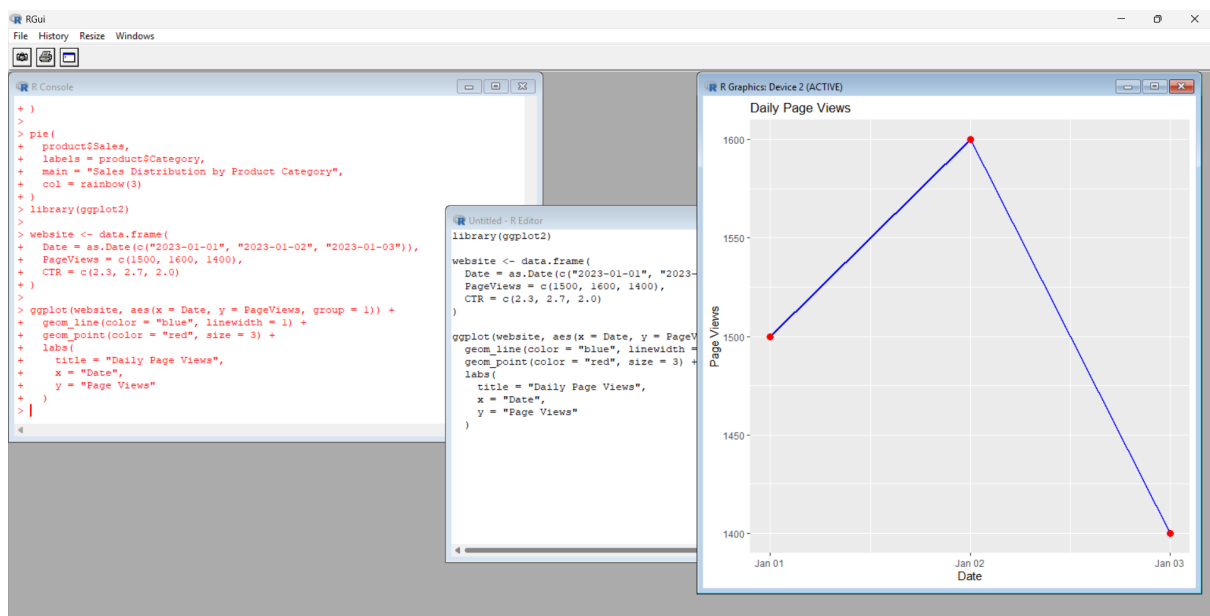
Set 12 :Dataset: Website Traffic

12A. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
library(ggplot2)
```

```
website <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),  
  PageViews = c(1500, 1600, 1400),  
  CTR = c(2.3, 2.7, 2.0)  
)
```

```
ggplot(website, aes(x = Date, y = PageViews, group = 1)) +  
  geom_line(color = "blue", linewidth = 1) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Daily Page Views",  
    x = "Date",  
    y = "Page Views"  
  )
```

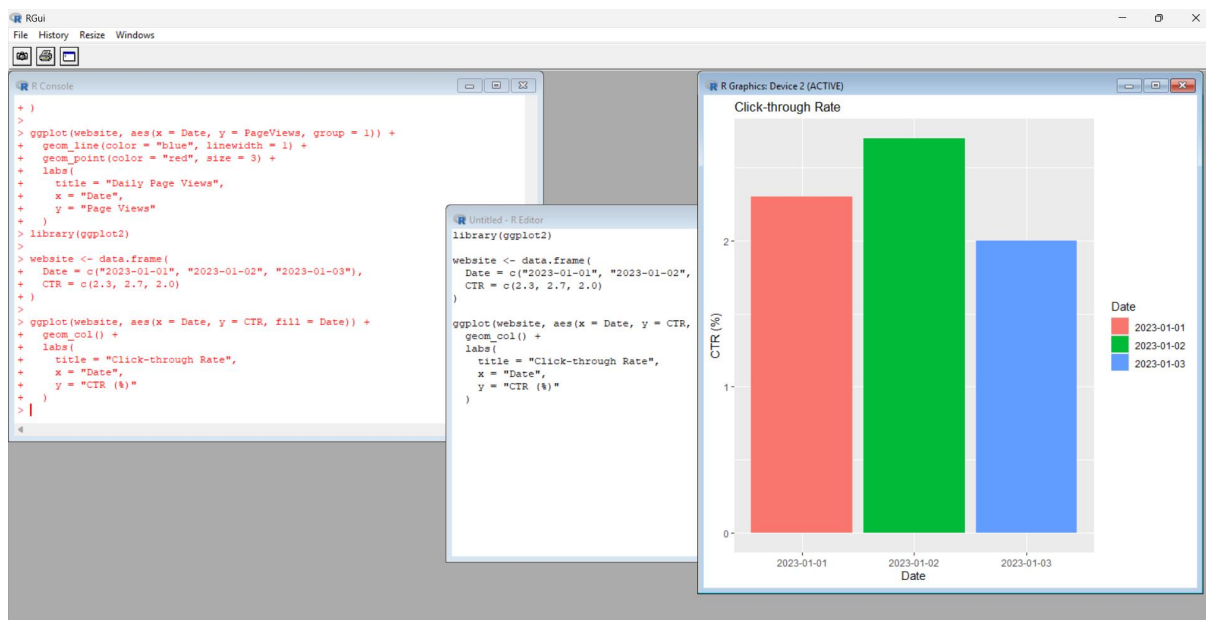


12B. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
library(ggplot2)
```

```
website <- data.frame(  
  Date = c("2023-01-01", "2023-01-02", "2023-01-03"),  
  CTR = c(2.3, 2.7, 2.0)  
)
```

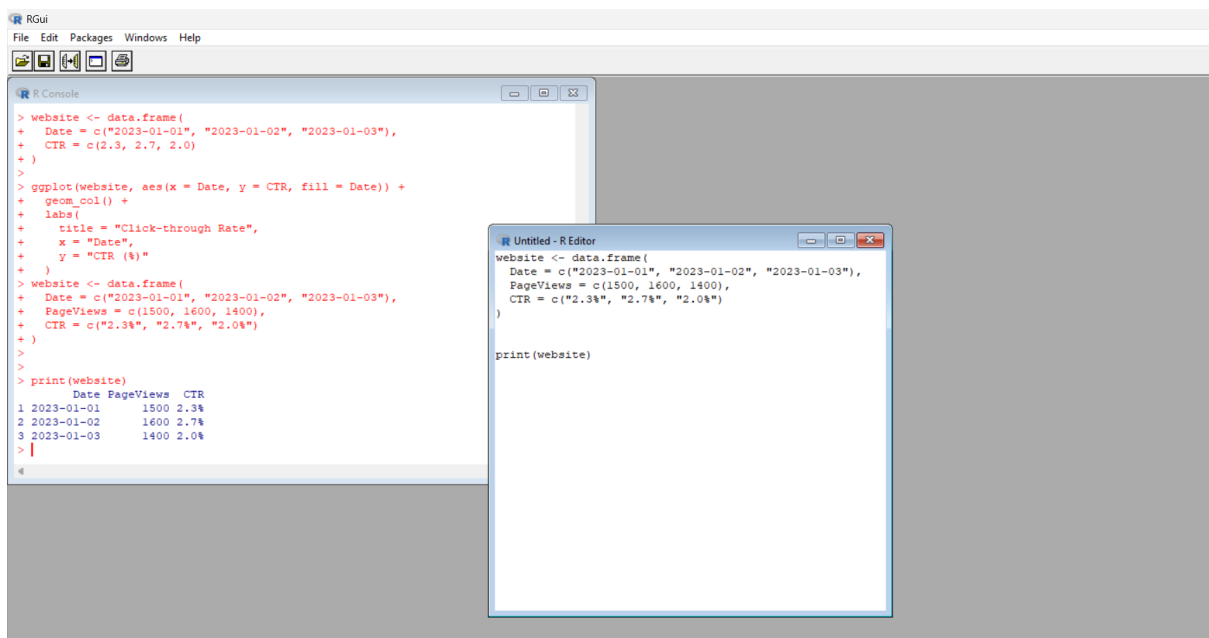
```
ggplot(website, aes(x = Date, y = CTR, fill = Date)) +  
  geom_col() +  
  labs(  
    title = "Click-through Rate",  
    x = "Date",  
    y = "CTR (%)"  
  )
```



12C. Build a table to show the website traffic data for each day. Label the table elements.

```
website <- data.frame(  
  Date = c("2023-01-01", "2023-01-02", "2023-01-03"),  
  PageViews = c(1500, 1600, 1400),  
  CTR = c("2.3%", "2.7%", "2.0%")  
)
```

```
print(website)
```



12D. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

```
library(ggplot2)
```

```
library(plotly)
```

```
website <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),  
  PageViews = c(1500, 1600, 1400),  
  CTR = c(2.3, 2.7, 2.0)  
)
```

```
p1 <- ggplot(website, aes(x = Date, y = PageViews, group = 1)) +  
  geom_line(color = "blue") +  
  geom_point(color = "red", size = 3)
```

```
ggplotly(p1)
```

```
p2 <- ggplot(website, aes(x = Date, y = CTR, fill = Date)) +  
  geom_col()
```

```
ggplotly(p2)
```

```
print(website)
```

